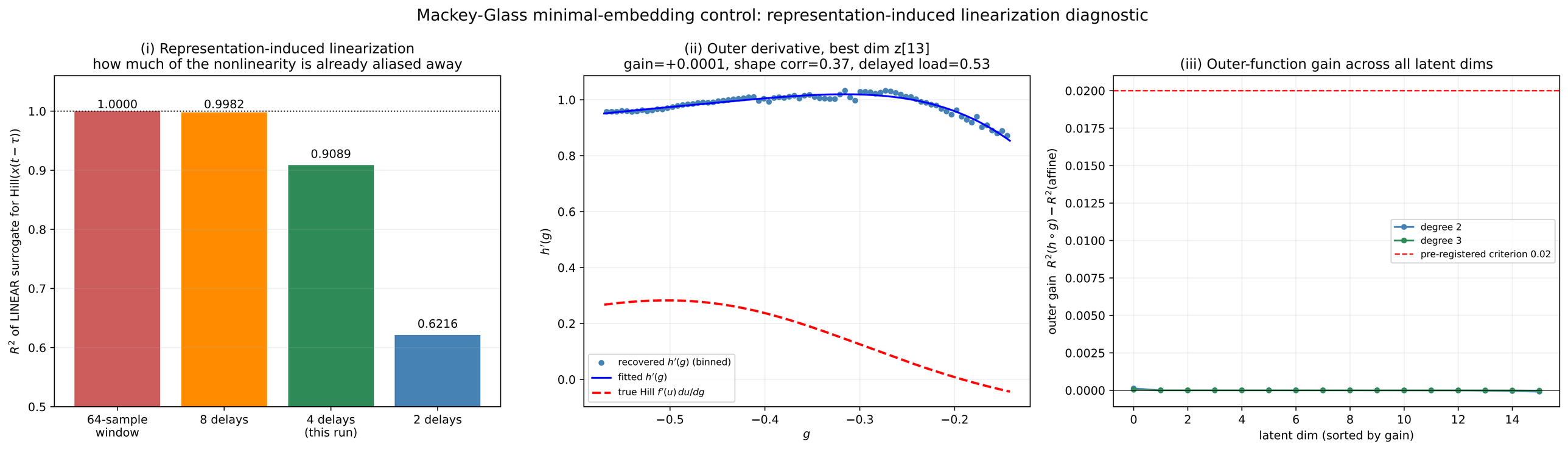
Mackey-Glass minimal-embedding control: representation-induced linearization diagnostic
(i) Representation-induced linearization
how much of the nonlinearity is already aliased away
R^2 of LINEAR surrogate for Hill($x(t-\tau)$)
1.0000
0.9982
0.9089
0.6216
64-sample window
8 delays
4 delays (this run)
2 delays
(ii) Outer derivative, best dim z[13]
gain=+0.0001, shape corr=0.37, delayed load=0.53
$h'(g)$
g
recovered $h'(g)$ (binned)
fitted $h'(g)$
true Hill $f'(u)\,du/dg$
(iii) Outer-function gain across all latent dims
outer gain $R^2(h \circ g) - R^2$(affine)
latent dim (sorted by gain)
degree 2
degree 3
pre-registered criterion 0.02